

Here's a concise **Quick Troubleshooting flow** suitable for Field Team as a guide

Quick Troubleshooting Flow – “No Connection to Unit” on Radio

A. No Connection to Unit



1. Check Power

- Is the unit powered ON?
- Check DC input, Fuse/MCB, and Power LED.
- **No Power → Restore power → Check if the Module is restored.**



2. Check Physical Connections

- Verify RF/coaxial cable and connectors.
- Check Ethernet/communication cable, if applicable.
- Look for loose/damaged connectors or cable cuts.
- **Connection issue → Rectify → Power cycle and ensure connectivity.**



3. Check Radio/Network Status

- Verify radio link/communication LEDs.
- Check signal/link status.
- Confirm correct frequency/channel/configuration.
- **Abnormal → Correct configuration/link → Re-test**



4. Check Unit Reachability

- Verify IP/address or unit ID.
- Perform ping/connectivity test where applicable.
- Check whether the unit is reachable from the management system.
- **Not reachable → Check transport/network path**



5. Check Alarms / PGOOD

- Review active alarms and event logs.
- Check **PGOOD-related alarms** as a unique fault cause.
- Verify whether the unit is repeatedly rebooting or losing power.



6. Restart / Reset

- Perform controlled power restart.
- Re-establish connection and monitor stability.



7. Still No Connection?

- Swap/replace suspected communication or RF hardware **only after confirming the fault.**
- Capture alarm, logs, measurements and replacement reasons.

On the BB module side:

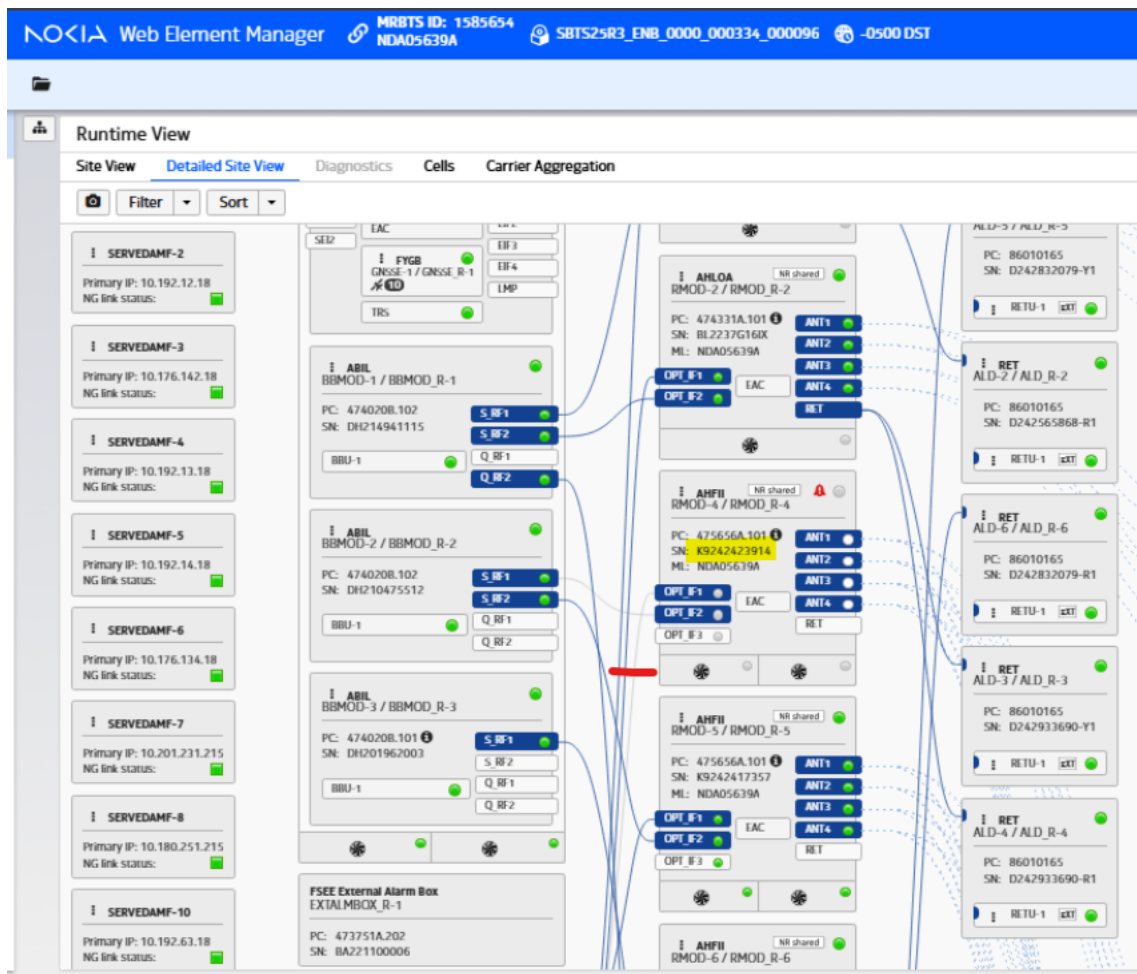
- 1 Fetch Snapshot
- 2 Ensure alarm with Fault ID 10 is still active on WebEM
- 3 Ensure the SFP(s) are securely plugged in
- 4 Ensure the fiber cables are securely plugged into SFPs
- 5 Clean SFP eyes and fiber cable ends
- 6 Give radio power reset to see if the problem got solved

On the radio module side:

- 7 Check DC voltage going to radio
- 8 Check radio led status
- 9 Connect LMI card to laptop with USB and Ethernet cables
- 10 Connect LMI card to radio's LMI port
- 11 Establish serial and ssh connections to radio
- 12 Fetch radio logs via ssh
- 13 Inspect optical fibers and connections visually
- 14 Ensure the SFP(s) are securely plugged in
- 15 Ensure the fiber cables are securely plugged into SFPs
- 16 Disconnect fiber cables from SFPs
- 17 Inspect SFP eyes and fiber cable ends with microscope, take photos

- 18 Check the SFP type and ensure its correct
- 19 Clean SFP eyes and fiber cable ends
- 20 Insert SFPs back to radio

1. Check if the connector seal is properly mounted and firmly in place. Plug out the SFPs, clean up or change of SFP/Fiber if needed.
2. Check if the SFP is properly connected and/or mounted in the RF module. If the SFP connection and/or mounting isn't correct, readjust the connection and/or mounting.
3. Check if the LED indicates the faulty SFP (connected to the system module). Replace the SFP if it's faulty.
4. Check if the optical cable is properly connected and/or mounted in the SFP.
5. Clean the optical connectors at the RF module end. Reset/Power cycle the RF HW unit and check if this solves the problem.
6. After all these steps if issue persist, Replace the Radio.



SFP Monitoring screenshot taken from WebEM , no readings available to view between the BBMOD_R-2(Left_Middle)/s_rf1/RMOD_R-4 serial_no=K9242423914

SFP Monitoring

SFPs applied to both near and far end in current BTS												
Link ports	Link speed	Fiber length	TX supply voltage	TX bias current	TX optical power(dBm)	TX optical power(W)	RX optical power(dBm)	RX optical power(W)	LCV	BER	Temperature	
RMOD-1/ABLOA/OPT_R2	10G0 Mb/s	109.63 m	3209 mV	15 mA	-2.322	579.1	-2.274	592.4	0	+	0	31 °C
CABINET-1/BBMOD-1/ABLU_S_RF1			3292 mV	12 mA	-1.058	494.5	-2.258	593.2	0	+	0	40 °C
RMOD-1/ABLOA/OPT_R2	10G0 Mb/s	109.21 m	3321 mV	16 mA	-2.384	577.6	-1.681	679.1	0	+	0	27 °C
CABINET-1/BBMOD-1/ABLU_S_RF2			3294 mV	15 mA	-3.41	456	-1.398	724.8	0	+	0	35 °C
RMOD-1/ABLOA/OPT_R2	10G0 Mb/s	109.7 m	3328 mV	15 mA	-2.109	615.3	-2.037	625.6	0	+	0	27 °C
CABINET-1/BBMOD-1/ABLU_S_RF2			3289 mV	14 mA	-1.87	680.8	-3.817	415.2	0	+	0	31 °C
				-	-	-	-	-	-	-		
				-	-	-	-	-	-	-		
				-	-	-	-	-	-	-		
RMOD-1/ABLOA/OPT_R2	10G0 Mb/s	109.21 m	3335 mV	13 mA	-2.074	620.3	-2.754	530.4	0	+	0	37 °C
CABINET-1/BBMOD-1/ABLU_S_RF2			3266 mV	13 mA	-1.166	764.6	-1.667	681.2	0	+	0	40 °C
RMOD-1/ABLOA/OPT_R2	10G0 Mb/s	109.7 m	3331 mV	11 mA	-1.816	608.2	-2.315	586.8	0	+	0	36 °C
CABINET-1/BBMOD-1/ABLU_S_RF1			3301 mV	15 mA	-1.718	664.1	-3.22	476.4	0	+	0	38 °C

SFPs applied to only one end in current BTS											
Ports	TX supply voltage	TX bias current	TX optical power(dBm)	TX optical power(W)	RX optical power(dBm)	RX optical power(W)	LCV	BER	Temperature		
CABINET-1/BBMOD-1/ABLU_S_RF1	3272 mV	14 mA	-1.709	674.7	Not determined	0	-	-	38 °C		
RMOD-1/ABLOA/OPT_R1	3316 mV	14 mA	-1.315	705.5	-1.838	654.9	0	+	0	33 °C	
RMOD-1/ABLOA/OPT_R1	3316 mV	14 mA	-2.797	525.2	-1.632	686.7	0	+	0	27 °C	
RMOD-1/ABLOA/OPT_R1	3327 mV	14 mA	-1.823	642.3	-3.423	453.6	0	+	0	27 °C	
RMOD-1/ABLOA/OPT_R1	3327 mV	12 mA	-1.662	682.1	-1.206	757.6	0	+	0	37 °C	
RMOD-1/ABLOA/OPT_R1	3326 mV	-	-	-	-	-	0	+	-	19 °C	
RMOD-1/ABLOA/OPT_R1	3338 mV	13 mA	-1.428	719.7	-4.216	373.6	0	+	0	40 °C	

- **Verify Physical Connections:**

- Ensure all network cables (Ethernet, fiber, etc.) connected to the unit are securely plugged in at both ends.
- Check for any visible damage to the cables or connectors.
- Clean the SFP's and the ports (sometimes the dust/moisture also interferes the connections).
- Replace the SFP's between BBMOD and Radio.
- Swap the fiber connections between the 2 sectors to see if the issue follows.

- **Check Power Supply:**

- Confirm that the unit is powered on and receiving adequate power.
- Look for power indicator lights on the unit.
- Verify that the power source (e.g., power outlet, power distribution unit) is functional.
- Look for any status LEDs on the unit that might indicate a fault.

- **Restart the Unit/System:**

- As a last resort, if all other checks pass, try power cycling the unit. This can sometimes resolve transient communication issues.
- Consider restarting any intermediate network devices (e.g., switches, routers) if they are suspected to be part of the communication path.

- **Infra (Power Outages)/Environmental Issue (Cold Weather, Storm, Lightning strike etc.,)**

To maintain reliable backup power, efficient energy management, and proactive maintenance at sites to prevent outages, especially during extreme weather or grid instability.

Also, operator make sure sites are installed with lightning rods (or lightning conductors), which attract strikes and safely guide the current to the ground, and surge protectors (or

SPD), which **protect electrical appliances** from **Voltage spikes** have lightning protection system (grounding) to direct electricity into the ground.

➔ Example from a site, **5TC3052A** where the AEHC radio was replaced by **TMO FOPS** due to **FID-10** however the actual issue was caused by other module not by the radio.

Actual module was **Jumper up top and SFPs**. They suggestion here is to replace strategically starting with smaller units like **SFP/Fibers/Jumpers** etc Instead of replacing **BBMOD/RADIO** at very first place.

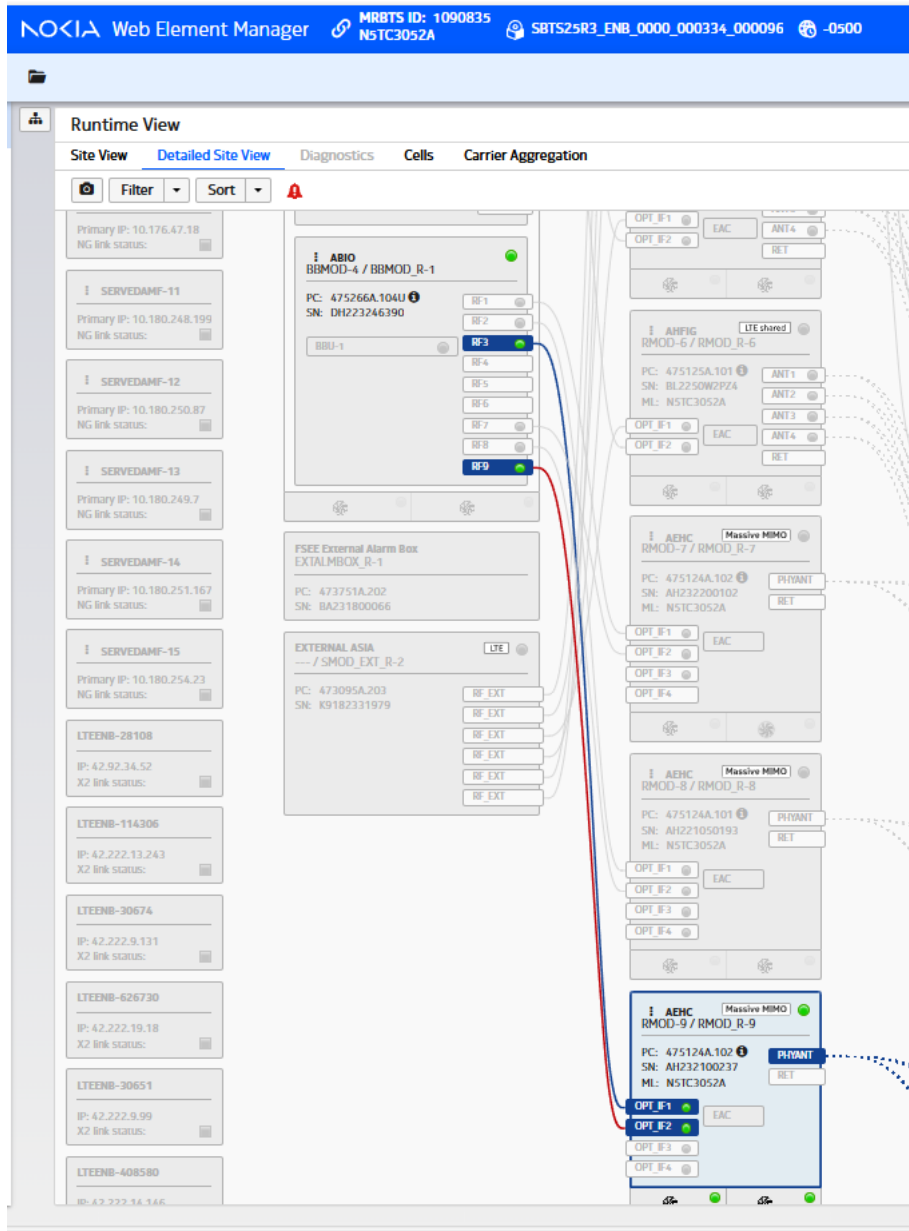
Also, we observed during this drill down exercise there were multiple site visits which also could be avoided.

Below are screenshots of the site during the recovery process where it went through different HW replacements: -

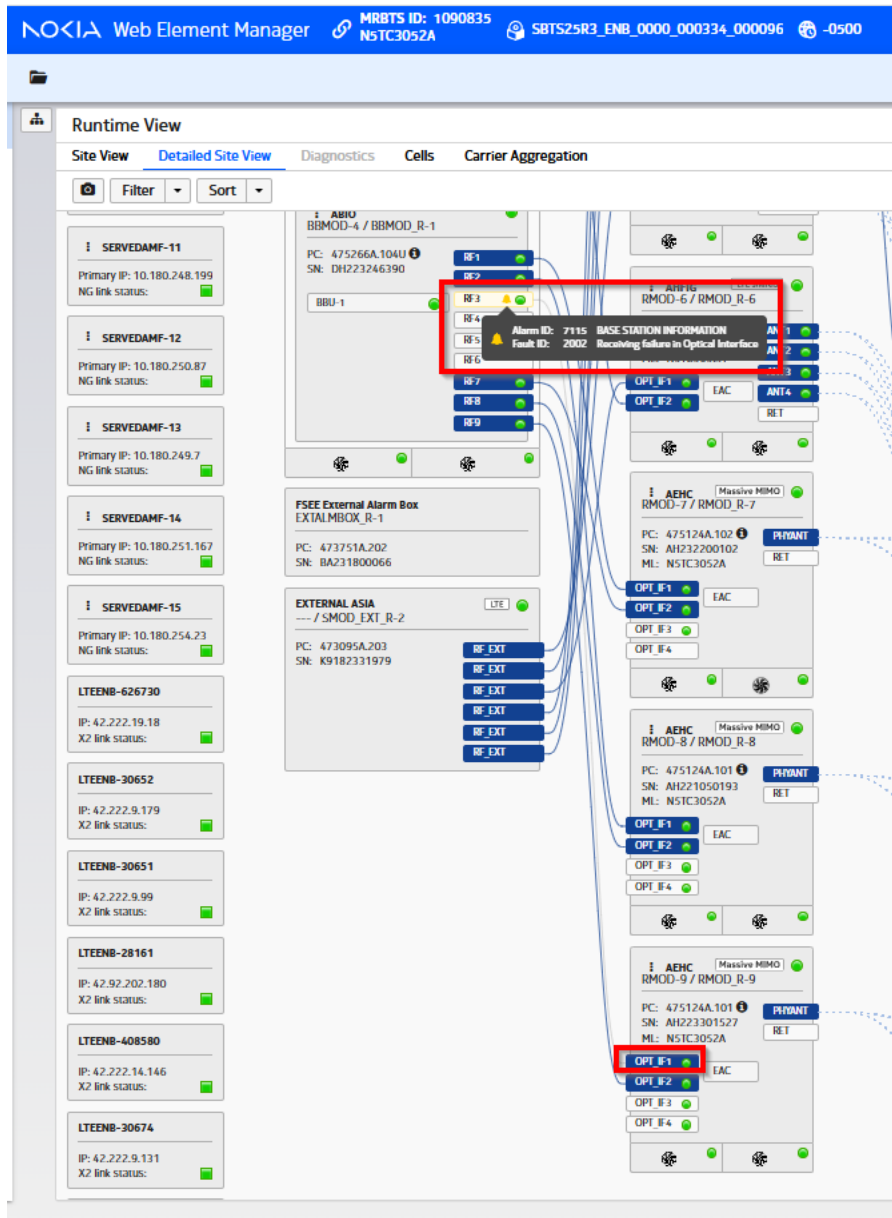
Replacement date – 06th March (Old serial- AH232100237 and New serial- AH223301527)

Supplementary Info	Diagnostic Info
I No connection to unit;Dedicated	10 unitName=AEHC path=/SMOD_R-1(Right)/BBMOD_R-1(Right_Bottom)/rf9/RMOD_R-9 serial_no=AH223301527 supplAlarmInfo=RU connection was lost in runtime :
I No connection to unit;Dedicated	10 unitName=AEHC path=/SMOD_R-1(Right)/BBMOD_R-1(Right_Bottom)/rf9/RMOD_R-9 serial_no=AH223301527 supplAlarmInfo=RU connection was lost in runtime :
Fronthaul loop back connectivity check failure	61009 supplAlarmInfo=Fronthaul loop back connectivity check failure on BB card MRBTS-1090835/EQM-1/APEQM-1/CABINET-1/BBMOD-4; interface RF9; vlan 6; level
Fronthaul loop back connectivity check failure	61009 supplAlarmInfo=Fronthaul loop back connectivity check failure on BB card MRBTS-1090835/EQM-1/APEQM-1/CABINET-1/BBMOD-4; interface RF9; vlan 6; level
I Unit power reset;Dedicated	4589 unitName=AEHC path=/SMOD_R-1(Right)/BBMOD_R-1(Right_Bottom)/rf9/RMOD_R-9 serial_no=AH223301527 additionalFaultID=4589
I No connection to unit;Dedicated	10 unitName=AEHC path=/SMOD_R-1(Right)/BBMOD_R-1(Right_Bottom)/rf9/RMOD_R-9 serial_no=AH223301527 supplAlarmInfo=RU connection was lost in runtime :
Fronthaul loop back connectivity check failure	61009 supplAlarmInfo=Fronthaul loop back connectivity check failure on BB card MRBTS-1090835/EQM-1/APEQM-1/CABINET-1/BBMOD-4; interface RF9; vlan 6; level
I Unit synchronization failed;Dedicated	16 unitName=AEHC path=/SMOD_R-1(Right)/BBMOD_R-1(Right_Bottom)/rf9/RMOD_R-9 serial_no=AH223301527 additionalFaultID=16
Fronthaul loop back connectivity check failure	61009 supplAlarmInfo=Fronthaul loop back connectivity check failure on BB card MRBTS-1090835/EQM-1/APEQM-1/CABINET-1/BBMOD-4; interface RF9; vlan 6; level
Fronthaul loop back connectivity check failure	61009 supplAlarmInfo=Fronthaul loop back connectivity check failure on BB card MRBTS-1090835/EQM-1/APEQM-1/CABINET-1/BBMOD-4; interface RF9; vlan 6; level
Fronthaul loop back connectivity check failure	61009 supplAlarmInfo=Fronthaul loop back connectivity check failure on BB card MRBTS-1090835/EQM-1/APEQM-1/CABINET-1/BBMOD-4; interface RF9; vlan 6; level
Fronthaul loop back connectivity check failure	61009 supplAlarmInfo=Fronthaul loop back connectivity check failure on BB card MRBTS-1090835/EQM-1/APEQM-1/CABINET-1/BBMOD-4; interface RF9; vlan 6; level
I Optical Interface faulty	2001 unitName=SFP path=/SMOD_R-1(Right)/BBMOD_R-1(Right_Bottom)/rf9/SFP_R-9 additionalFaultID=2001
I No connection to unit;Dedicated	10 unitName=AEHC path=/SMOD_R-1(Right)/BBMOD_R-1(Right_Bottom)/rf9/RMOD_R-9 serial_no=AH223301527 supplAlarmInfo=RU connection was lost in runtime :
I Optical Interface faulty	2001 unitName=SFP path=/SMOD_R-1(Right)/BBMOD_R-1(Right_Bottom)/rf9/SFP_R-9 additionalFaultID=2001
I Optical Interface faulty	2001 unitName=SFP path=/SMOD_R-1(Right)/BBMOD_R-1(Right_Bottom)/rf9/SFP_R-9 additionalFaultID=2001
I Receiving failure in Optical Interface	2002 unitName=LANE path=/SMOD_R-1(Right)/BBMOD_R-1(Right_Bottom)/rf9/SFP_R-9/LANE_R-1 additionalFaultID=2002
I Optical Interface faulty	2001 unitName=SFP path=/SMOD_R-1(Right)/BBMOD_R-1(Right_Bottom)/rf9/SFP_R-9 additionalFaultID=2001
I No connection to unit;Dedicated	10 unitName=AEHC path=/SMOD_R-1(Right)/BBMOD_R-1(Right_Bottom)/rf9/RMOD_R-9 serial_no=AH223301527 supplAlarmInfo=RU connection was lost in runtime :
I Optical Interface faulty	2001 unitName=SFP path=/SMOD_R-1(Right)/BBMOD_R-1(Right_Bottom)/rf9/SFP_R-9 additionalFaultID=2001
I Optical Interface faulty	2001 unitName=SFP path=/SMOD_R-1(Right)/BBMOD_R-1(Right_Bottom)/rf9/SFP_R-9 additionalFaultID=2001
I No connection to unit;Dedicated	10 unitName=AEHC path=/SMOD_R-1(Right)/BBMOD_R-1(Right_Bottom)/rf9/RMOD_R-9 serial_no=AH232100237 supplAlarmInfo=RU connection was lost in runtime :
I Receiving failure in Optical Interface	2002 unitName=LANE path=/SMOD_R-1(Right)/BBMOD_R-1(Right_Bottom)/rf9/SFP_R-9/LANE_R-1 additionalFaultID=2002
I Not able to synchronize baseband bus;Dedicated	1815 unitName=Cablink path=/SMOD_R-1(Right)/BBMOD_R-1(Right_Bottom)/rf9 supplAlarmInfo=The eCPRI synchronization is lost on cable between system module/
Fronthaul loop back connectivity check failure	61009 supplAlarmInfo=Fronthaul loop back connectivity check failure on BB card MRBTS-1090835/EQM-1/APEQM-1/CABINET-1/BBMOD-4; interface RF9; vlan 6; level

FID61009 - Fronthaul Loopback Fail on BBMOD-4 RF9(ABIO) to RMOD-9 IF-2(AEHC).



HARDWARE - ABIO-4 RF3 RECEIVING FAILURE IN OPTICAL INTERFACE / BTS CONFIGURATION ERROR (not enough HW for LCR) 1099 - 5TC3052A - BREAKFIX - HARDWARE - POSSIBLE ABIO BBMOD4 FAULTY ON RF3 PORT



After hours of troubleshooting and replacing ABIO I moved Fiber from RF9 to RF3 and confirmed port RF3 is good. Issue is up the Fiber path to the tower top.

This confirms Port RF3 and the ABIO, AMIA, and ASILs.

The screenshot displays a network management interface for a radio access network. The main area shows a 'Runtime View' with a 'Detailed Site View' tab selected. The site view includes a list of served areas (SERVEDAMF-10 to SERVEDAMF-13) and a central diagram of the network components. The diagram shows a BBU-1 connected to an ABIO (BBMOD-4 / BBMOD_R-1) and an AEHC (Massive MIMO RMOD-7 / RMOD_R-7). The ABIO has RF ports RF1 through RF9. The AEHC has OPT_IF1 through OPT_IF4. The diagram shows connections between the RF ports and the OPT_IF ports. The right-hand panel shows 'Details' for the site, including 'BTS site properties' and 'Additional information'.

BTS site properties

BTS name:	N5TC3052A
BTS ID:	1090835
Location:	5133 Kingsport Highway
Time zone:	(GMT-5) US/Eastern
Software version:	SBTS25R3_ENB_0000_000334_00096
Distinguished name prefix:	-
Managed element:	MRBTS-1090835

Additional information

BTS description	-
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I put the original ABIO back in and moved RF9 Fiber to RF3 and RF3 on this original ABIO is working. Another confirmation the issue appears to be in the Fiber path from RF3 to IF1 of gamma AEHC.

INC1245 x MRBTS x RIOT 2.0 x KX_WV x Open W x OneCon x Sign in x Sign out x + Ask Gemini

Not secure https://11.94.15.170/#/bts-status/detailed-site-view

Managed bookmarks Login | Axiom RIOT 2.0 My Workspace | Ser... Compliance Map -... One Console NEST Personal Work-Commonly U...

NOKIA Web Element Manager MRBTS ID: 1090835 N5TC3052A SBT525R3_ENB_0000_00... -0400 DST

On air NR Not commissioned LTE Not commissioned WCDMA

Runtime View

Site View Detailed Site View Diagnostics Cells Carrier Aggregation

Filter Sort

SERVEDAMF-10

Primary IP: 10.176.47.18
NG link status: ■

SERVEDAMF-11

Primary IP: 10.180.248.199
NG link status: ■

SERVEDAMF-12

Primary IP: 10.180.250.87
NG link status: ■

SERVEDAMF-13

Primary IP: 10.180.249.7
NG link status: ■

ABIO

BBMOD-4 / BBMOD_R-1
PC: 475266A.104U
SN: DH223246390
BBU-1

RF1 RF2 RF3 RF4 RF5 RF6 RF7 RF8 RF9

AEHC

Massive MIMO
RMOD-7 / RMOD_R-7
PC: 475124A.102
SN: AH232200102
ML: N5TC3052A

PHYANT

OPT_IF1 OPT_IF2 OPT_IF3 OPT_IF4

475124A.102

Details

Info Parameters Faults

BTS site properties

BTS name:	N5TC3052A
BTS ID:	1090835
Location:	5133 Kingsport Highway
Time zone:	(GMT-5) US/Eastern
Software version:	SBT525R3_ENB_0000_00096
Distinguished name prefix:	-
Managed element:	MRBTS-1090835

Additional information

BTS description

After previous topside investigation and extensive downstairs troubleshooting by FOPs, the issue appears to be in the fiber path from baseband card RF3 to IF1 of gamma AE

HC. Need more topside investigation.



Resolution Notes: -

TC replaced jumper up top and replaced SFP at radio. Also corrected a few other minor issues while onsite.

B. Troubleshooting Fault ID 1907 TX out of order

START → Fault 1907 “TX Out of Order”



1. Check Alarm Status

- Confirm Fault ID **1907** is active.
- Check whether the alarm is **intermittent or persistent**.



2. Check Radio Power

- Verify unit/Radio power is stable.
- Check **DC input, fuse/MCB and power LEDs**.
- Look for repeated reboot/power fluctuation.

Power abnormal → Rectify power issue → Clear alarm → Re-test



3. Check TX Path

- Inspect TX/RF cable and connectors.
- Check for loose, damaged or improperly seated connectors.
- Verify RF path/feeder condition.
- Check for abnormal VSWR/reflected power, if available.

TX path abnormal → Rectify → Re-test



4. Check Radio Configuration

- Verify TX frequency/channel configuration.
- Check TX enable/mute status.
- Confirm configuration matches the connected unit/network.
- Check recent configuration changes.



5. Check TX Hardware Status

- Review radio/module diagnostics.
- Check **PA/TX module alarms** and temperature status.
- Check whether TX output is actually being generated.

TX output absent/abnormal → Suspect TX/PA hardware



6. Check Related Alarms

Look for accompanying alarms such as:

- PA/Power Amplifier fault
- High VSWR
- Over-temperature
- PGOOD/power-related alarm
- Module communication failure
- PLL/frequency-related fault



7. Reset & Re-test

- Perform controlled reset/restart as per approved procedure.
- Clear the alarm.
- Verify TX operation and monitor for recurrence.



If Fault 1907 Persists

Confirm external RF path → Confirm configuration → Check TX/PA module → Replace HW only after fault confirmation → Escalate with logs/alarms and test results.

Recommended fault classification:

1907 TX Out of Order → TX/Radio Transmission Path Fault

If investigation confirms a **PA/TX hardware failure**, classify the **unique cause as TX/PA Hardware Fault** rather than simply “TX Out of Order.”

C. Nokia Fault 1815 – (Not able to synchronize baseband bus)

eNodeB (LTE) or gNodeB (NR) cannot establish **Synchronization** between the **System Module (ASIL/ASIM/BBMOD)** and the **RF Module / Radio Module** over the Baseband Bus (**BB Bus / RP3 interface**). This usually **causes cells** or the whole BTS to go out of service (OoS).

Fault ID: **1815**

- **1815: EFaultId_BbBus_SyncFailedAl**

Typical Alarms reported:

- 7650 Base Station Faulty
- 7651 Base Station Operation Degraded
- 7653 Cell Faulty
- 7654 Cell Operation Degraded

Main Causes of Fault 1815

1. BB Bus / Optical Cable Problem

Most common reason:

- Loose or damaged CPRI/RP3/OBSAI cable
- Wrong cable connection
- Dirty optical connector
- Fiber bend or attenuation issue

2. RF Module Failure

The RF module may:

- Hang during startup
- Fail synchronization
- Have internal FPGA/DSP failure
- Overheat or reset repeatedly

3. System Module Failure

Possible issues:

- Faulty FSP/ASIL/ASIM/BBMOD
- Corrupted synchronization configuration
- HW communication failure

4. Software or Configuration Mismatch

- Wrong BTS configuration
- Incorrect RP3 link speed
- Incompatible SW versions
- Incomplete commissioning

5. Clock / Sync Issue

If reference clock is unstable:

- BB bus synchronization may fail
- Related alarms like 1816 may appear

Troubleshooting Procedure (Recommended by Nokia Practice)

Step 1 — Check Other Active Alarms

First verify if any of these faults are in the Active Alarm or History:

- 1811
- 1904
- 1905
- 1906
- No connection to unit alarm (10)

These often indicate the real root cause.

Step 2 — Check Physical Connectivity

Inspect carefully:

- Fiber/copper BB bus cable
- CPRI cable seating
- Optical SFPs
- RF module ports
- Connector cleanliness

Actions:

- Remove and reconnect
 - Swap fiber
 - Clean optical ports
 - Verify TX/RX not crossed
-

Step 3 — Verify Module LEDs

Check:

- RF module LEDs
- System module LEDs

Typical indication:

- Red/blinking red(HW or sync failure)
 - No LED (power or hardware issue)
-

Step 4 — Reset the Faulty Unit

Recommended recovery:

1. Block RF module
2. Unblock RF module
3. If not recovered:
 - Block/unblock System Module

4. Reboot BTS if necessary
-

Step 5 — Check BB Bus Configuration

Verify:

- RP3 link speed
- Radio type mapping
- Sector-to-radio mapping
- Daisy chain/routing direction

Wrong routing direction can trigger BB synchronization alarms.

Step 6 — Swap Hardware

If fault persists:

1. Replace BB bus cable
 2. Replace SFPs at both the ends (BBMOD & RMOD)
 3. Replace RF module
 4. Replace System Module
-

Field Experience Tips

Common practical observations from field engineers:

- **70% cases** (Bad Fiber/CPRI connection)
 - Frequent after power fluctuation
 - Common after incorrect commissioning
 - Sometimes caused by incompatible radio software
 - Water ingress in outdoor RF modules can trigger 1815 intermittently.
-

Quick Troubleshooting Flow

Check for all service impacting Alarms (Active and Historical as well – **15 Days**).



Check/Inspect **Fiber/BB cable**



Check RF module LEDs



Block/Unblock RF module.



Power cycle the corresponding BBMOD/RMOD



Restart BTS



Verify **RP3/CPRI** configuration



Swap Cable/SFPs



Replace RF module



Replace System module

Important to note these Related Faults like FID1815:

Fault ID	Meaning
1811	BB Bus configuration rejected

Fault ID	Meaning
1904	RF BB Bus synchronization error
1905	RF BB Bus configuration error
1906	RF BB Bus transmission error
3020	Baseband Bus failure
3080	BTS reference clock missing

D. Troubleshooting VSWR and RTWP Issues

VSWR and **RTWP** are among the most common in **LTE/5G RF** issues affecting **uplink/downlink quality, call drops, low throughput, and coverage degradation**.

1. VSWR Issue Troubleshooting

What is VSWR?

VSWR (Voltage Standing Wave Ratio) indicates impedance mismatch between:

- RRU/RRH
- Jumper
- Feeder cable
- Antenna

High VSWR means reflected power is returning to the Radio.

Normal:

- ≤ 1.5 : **Good**
 - 1.5–2.0 : **Acceptable**
 - 2.0 : **Problematic**
-

Symptoms of High VSWR

- Cell degraded alarm
- Low DL coverage
- High reflected power
- RRU PA shutdown
- Call drops
- Low throughput
- Sector outage in severe cases

Typical alarms:

- VSWR threshold crossed
 - TX path fault
 - Antenna line fault
 - High reflected power alarm
-

Main Causes of VSWR

Cause	Description
Loose jumper	Connector not tightened
Damaged jumper	Water ingress or bent cable
Faulty feeder	Internal cable damage
Bad antenna	Internal antenna failure
Water ingress	Moisture in connector

Cause	Description
Corroded connector	Oxidation increases mismatch
Wrong connector type	DIN/N-type mismatch
Lightning damage	Burns RF path
Poor installation	Improper connector crimping

1. VSWR Troubleshooting Procedure

Step 1 — Check Alarms

Verify:

- Which sector?
- Which TX/RX branch?
- Intermittent or permanent?

Check:

- VSWR value
 - Reflected power
 - TX power reduction
-

Step 2 — Physical Inspection

Inspect:

- Jumper cables
- Feeder connectors
- Antenna ports
- Weatherproofing tape
- Water leakage

Look for:

- Rust
 - Blackened connectors
 - Moisture
 - Loose DIN connectors
-

Step 3 — Tighten Connectors

Use proper torque wrench:

- Over-tightening also causes problems.

Common field issue:

- Connector finger-tight only.
-

Step 4 — Swap Jumpers

Swap:

- Jumper with healthy sector

If fault moves:

→ Jumper faulty.

Step 5 — Measure with Site Master / Anritsu

Best practice:

- Disconnect antenna line
- Measure Return Loss / VSWR

Typical findings:

- Distance-to-fault (DTF)
- Connector mismatch
- Feeder damage point

Step 6 — Check Antenna

Possible issues:

- Internal antenna damage
- Water ingress
- Burned diplexer
- PIM issue

Swap antenna port if possible.

Step 7 — Check RRU/RRH Port

Sometimes issue is:

- Damaged RF port
- Burned circulator
- Internal PA failure

Swap RF port/branch.

VSWR Quick Isolation Logic

Alarm appears

↓

Check connector tightness

↓

Swap jumper

↓

Measure feeder with Site Master

↓

Replace jumper/connector

↓

Check antenna

↓

Check RRU port

2. RTWP Issue Troubleshooting

What is RTWP?

RTWP (Received Total Wideband Power) measures total uplink noise/interference at the receiver.

Used mainly in:

- WCDMA
- LTE uplink analysis

Normal RTWP:

- Around -105 dBm to -95 dBm (depends on load)

Problem indication:

- One branch significantly higher than others
-

Symptoms of RTWP Issue

- UL interference
- Poor upload throughput
- Call setup failure
- High BLER
- Accessibility degradation
- UL packet loss

Typical KPI impact:

- CSSR degradation

- Drop increase
- UL throughput low

Main Causes of RTWP Imbalance

Cause	Description
External interference	Noise source nearby
Faulty TMA	Generates noise
Water ingress	Creates passive intermodulation
Bad jumper	UL noise increase
Faulty antenna	Internal imbalance
PIM issue	Passive Intermodulation
Cross feeder issue	Improper cabling
RRU receiver fault	Defective RX chain

RTWP Troubleshooting Procedure

Step 1 — Compare RTWP Between Branches

Healthy:

Branch A = -101 dBm

Branch B = -102 dBm

Problem:

Branch A = -88 dBm

Branch B = -102 dBm

Difference > 3 dB usually indicates issue.

Step 2 — Check Time Pattern

Questions:

- Constant issue?
- Busy hour only?
- Random spikes?

If busy-hour only:

→ External interference likely.

Step 3 — Swap RX Paths

Swap:

- Jumper
- RX branch
- Antenna port

If issue moves:

→ External RF path issue.

If stays:

→ RRU issue.

Step 4 — Check PIM Sources

Common causes:

- Loose connector
- Rusted feeder
- Metal contact near antenna
- Poor grounding

PIM creates uplink noise.

Step 5 — Spectrum Analysis

Use:

- Spectrum analyzer
- Interference hunting tools

Look for:

- Repeater interference
- Illegal booster
- External transmitter

Step 6 — Disable External Devices

Check:

- TMA
- RET
- Combiner
- Diplexer

Bypass temporarily.

Step 7 — Check Antenna Alignment

Wrong tilt or damaged antenna can:

- Increase interference pickup
- Cause RTWP imbalance

RTWP Isolation Logic

High RTWP detected

↓

Compare branches

↓

Swap RX paths

↓

Check jumper/feeder

↓

Check PIM sources

↓

Spectrum analysis

↓

Replace antenna/TMA

↓

Replace RRU

Difference Between VSWR and RTWP

Parameter	VSWR	RTWP
Direction	Downlink TX path	Uplink RX path
Main issue	Reflection	Interference/noise
Affects	TX efficiency	UL sensitivity
Common tool	Site Master	Spectrum Analyzer
Main symptom	Reflected power	High UL noise

Practical Field Tips

VSWR

- Most common cause = loose jumper
- Water ingress is extremely common in coastal areas
- Always inspect weatherproofing

RTWP

- PIM is often hidden root cause
- Compare neighboring sectors
- Interference may come from external operators

Important KPI Thresholds

Parameter	Healthy
VSWR	< 1.5
Return Loss	> 14 dB
RTWP imbalance	< 3 dB
UL Noise Floor	Around -100 dBm

➤ One of the examples of these issues we have witnessed recently where **HW (Radio) is not replaced.**

BTS Name: **M9BH8040A**

MRBTS ID: **1113869**

HW Module/SN: **AHLOB/ AH252321874**

Alarms per NAF History: VSWR (**1837**) / Transmission path failure (4003) / Antenna line switched off due to high VSWR (**1957**).

TMO INC's (Tickets): **INC125072668** (Closed) / **INC126848908** (Open)

Radio Replacement status: **Radio not replaced.**

Issue: **VSWR**

Resolution: Tower crew to troubleshoot Alpha AHLOB SN:AH252321874 VSWR upper threshold exceeded -ANT2. FOPS incident **TC replaced the Jumper.**

Replaced jumper. Troubleshooting the Alpha AHLOB (S# AH252321874) for a Major VSWR alarm. **Tested the RF Jumpers** and **cleaned the connections.**

Replaced the RF Jumper on the Alpha AHLOB. Issue resolved. Logged out with Semester Grimes. Site complete.

- E. **NokiaFault-1927_RF_Power_decreased** indicates a detected reduction in the Radio Frequency (RF) output power from a Nokia network element, which can impact network performance and coverage.

Understanding NokiaFault-1927_RF_Power_decreased

This fault signifies that the RF power output from a base station or radio unit has fallen below its expected or configured level. This can lead to reduced cell coverage, lower signal quality for users, and decreased data throughput.

Common Causes for Decreased RF Power in Nokia Equipment

Decreased RF power can stem from various issues, including:

- **Hardware Malfunctions:**
 - **Faulty Radio Unit (RRH/RFM):** The radio unit itself might be defective.
 - **Antenna Issues:** Damaged or misaligned antennas, or issues with the antenna's internal components.
 - **Feeder Cable Problems:** Damaged, corroded, or improperly connected RF feeder cables (coaxial cables) between the radio unit and the antenna.
 - **Connector Issues:** Loose, corroded, or damaged RF connectors.
 - **Power Supply Problems:** Insufficient or unstable power supply to the radio unit.
- **Configuration Errors:**
 - **Incorrect Power Settings:** RF power levels might be misconfigured in the network management system or directly on the equipment.
 - **Software Glitches:** Bugs or errors in the radio unit's software.
- **Environmental Factors:**
 - **Extreme Temperatures:** Operating outside specified temperature ranges can affect RF component performance.
 - **Moisture Ingress:** Water damage to cables, connectors, or radio units.
- **External Interference:** While less common as a direct cause of decreased output power, severe interference can sometimes be misinterpreted or contribute to system instability that affects power output.

Troubleshooting Steps for NokiaFault-1927_RF_Power_decreased

To diagnose and resolve this fault, follow these general troubleshooting steps:

1. Check Alarms and Logs:

- Verify if other related alarms are present (e.g., VSWR alarms, hardware alarms, power supply alarms).
- Review system logs for any events preceding the fault.

2. Physical Inspection:

- **Cables and Connectors:** Visually inspect all RF feeder cables and connectors for damage, corrosion, or loose connections. Ensure proper grounding.
- **Antenna:** Check the antenna for physical damage or misalignment.
- **Radio Unit:** Inspect the radio unit for any visible damage, proper seating, and ensure all cables are securely connected.

3. Verify Power Supply:

- Confirm that the radio unit is receiving stable and correct power voltage.

4. Configuration Review:

- Access the network management system (e.g., NetAct) or the local management interface of the radio unit.
- Verify that the RF power settings are correctly configured according to the network plan.
- Check for any recent configuration changes that might have triggered the issue.

5. Perform VSWR Measurement:

- If possible, perform a Voltage Standing Wave Ratio (VSWR) measurement to check the impedance match of the antenna system. High VSWR indicates a problem with the antenna, feeder cable, or connectors.

6. Software Check:

- Ensure the radio unit is running the approved software version. Consider a software upgrade or downgrade if recent changes correlate with the fault.

7. Component Swap (Last Resort):

- If all other steps fail, consider swapping the suspected faulty component (e.g., radio unit, feeder cable, antenna) with a known good one to isolate the problem.

Troubleshooting environmental external alarms focuses on the specific sensors and conditions they monitor. Here's a systematic approach:

Troubleshooting Environmental External Alarms

1. Identify the Specific Environmental Alarm:

- Determine the exact type of environmental alarm (e.g., high temperature, low temperature, high humidity, water leak, smoke/fire, air quality, power failure).
- Note the alarm's exact message, threshold, and the sensor ID or location from your monitoring system.

2. Verify Environmental Conditions:

- **Direct Measurement:** Use a calibrated, independent device (e.g., thermometer, hygrometer, water detection strip) to physically verify the environmental condition at the sensor's location.
 - Example: If a high-temperature alarm triggers, check the actual temperature with a handheld thermometer.
- **Visual Inspection:** Look for obvious issues in the environment:
 - **Temperature:** Blocked air vents, malfunctioning HVAC, open doors, heat-generating equipment.
 - **Humidity:** Leaks, condensation, humidifier/dehumidifier issues.
 - **Water Leak:** Visible water, drips, condensation, burst pipes.
 - **Smoke/Fire:** Visible smoke, burning smell, heat.

3. Inspect the Sensor:

- **Physical Condition:** Check the sensor for damage, dirt, dust, or obstructions that might affect its readings. Ensure it's securely mounted and positioned correctly.
- **Placement:** Verify the sensor is in an appropriate location to accurately monitor the desired condition (e.g., not directly in front of an air vent, away from direct sunlight).
- **Connectivity:** Ensure all cables connecting the sensor to the alarm panel or monitoring unit are secure, undamaged, and properly terminated.

4. Check Sensor Calibration and Thresholds:

- **Calibration:** If the sensor is field-calibratable, check its last calibration date and consider recalibration if readings are consistently inaccurate compared to your independent measurements.
- **Alarm Thresholds:** Review the configured alarm thresholds in the monitoring system. Ensure they are set appropriately for the environment and that the current (verified) conditions are not legitimately exceeding them.

5. Power and Communication:

- **Sensor Power:** Confirm the sensor is receiving power (if applicable). Check indicator lights on the sensor or its associated module.
- **Communication Link:** Verify the communication path between the sensor and the monitoring system (e.g., wired connection, wireless signal strength, network connectivity).

6. Review Logs and History:

- Check the monitoring system's event logs for a history of the alarm. Look for patterns, intermittent triggers, or other related alarms that might provide context.

7. Test the Sensor (if safe and feasible):

- **Temperature:** Gently apply a heat source (e.g., hairdryer on low, but not directly on the sensor) or a cold source (e.g., ice pack wrapped in cloth) to see if the sensor responds and triggers the alarm at the set threshold.
- **Water Leak:** Apply a small amount of water to the water leak sensor.
- **Smoke:** Use approved test smoke spray for smoke detectors.

8. Reset and Monitor:

- After addressing any identified issues, clear the alarm (if possible) and monitor the system to confirm the alarm does not reoccur.